

Keywords: diffusion language models • reinforcement learning • arbitrary-order generation • reasoning

The Flexibility Trap

Arbitrary-Order Generation Narrows, Not Expands, Reasoning in Diffusion Language Models

By Zanolin Ni, Shenzhi Wang, Yang Yue, Tianyu Yu, Weilin Zhao, Yeguo Hua, Tianyi Chen, Jun Song, Cheng Yu, Bo Zheng & Gao Huang (Tsinghua University et al.)

arXiv:2601.15165 • ICML 2026 Outstanding Paper Award

Diffusion large language models promise to supersede autoregressive generation by producing tokens in any order, not just left-to-right. The promise has a flaw: for reasoning tasks like mathematics and coding, that same flexibility teaches models to *avoid* uncertain tokens rather than resolve them, collapsing the solution space before it has been explored. A minimalist fix—JustGRPO, which constrains rollouts to left-to-right during RL training only—achieves **89.1%** on GSM8K, outperforming every prior diffusion RL method by more than six percentage points.

AT A GLANCE

Problem: Diffusion LLMs exploit arbitrary-order generation to skip high-uncertainty tokens during RL exploration, collapsing solution coverage.

Why it matters: dLLMs are positioned as the next generation of language model architectures; if their defining feature sabotages reasoning, that claim requires fundamental reassessment.

Method: Coverage analysis of rollout distributions; JustGRPO constrains exploration to left-to-right ordering while leaving inference unchanged.

Result: 89.1% on GSM8K and 45.1% on MATH-500, surpassing best prior diffusion RL methods (82.8% and 39.6%) by large margins.

Evidence: Comparisons against ESPO and GDPO on GSM8K, MATH-500, and AIME; ablations confirm the constraint matters only at training time.

The Big Picture

Diffusion language models (dLLMs) generate text by iteratively denoising a masked token sequence, selecting which token to reveal at each step in any order. This departs from the strictly left-to-right generation of autoregressive transformers and, in theory, should allow dLLMs to explore a solution space that strictly subsumes the autoregressive one: any left-to-right trajectory is a special case of arbitrary-order generation.

Recent work applied reinforcement learning to dLLMs for reasoning tasks, producing methods like ESPO and GDPO. These inherit the arbitrary-order assumption uncritically. This paper asks a prior question: *is that assumption correct?*

The answer is no. Arbitrary-order generation, when combined with RL exploration, traps models in a degenerate strategy that undermines the very diversity RL is meant to exploit.

Why Existing Approaches Fall Short

ESPO and GDPO apply group relative policy optimization directly to dLLM rollouts, allowing arbitrary generation order throughout. The coverage analysis reveals the consequence: models learn to generate high-confidence (low-entropy) tokens first, committing early to a particular solution path, and defer high-uncertainty tokens to the end where their influence on the output is minimal.

This is structurally equivalent to a tree search algorithm that always expands its most confident node and never backtracks. The diversity in the rollout batch that GRPO needs for advantage estimation collapses: many rollouts converge to the same prefix, and the effective group size shrinks.

Prior interpretations attributed performance gaps to model size or training duration. The coverage diagnostic shows the bottleneck is the order constraint: the problem is present from the first RL iteration.

Core Mechanism

JustGRPO makes a single modification to the dLLM RL pipeline: during the rollout generation phase, tokens are produced strictly left-to-right. The model’s bidirectional attention and parallel decoding at inference are completely unchanged.

This forces the model to resolve uncertain tokens in context order during exploration, maintaining a diverse rollout distribution. The GRPO advantage estimator then has access to genuine diversity—rollouts that took genuinely different paths—producing meaningful gradient signal.

At inference, the model reverts to its native arbitrary-order parallel decoding, so the speed benefits of dLLMs are fully preserved.

Technical Formulation

Let G rollouts $\{o_1, \dots, o_G\}$ be sampled from the current policy π_θ for prompt q . Define normalized advantages

$$\hat{A}_i = \frac{r_i - \text{mean}(r_{1:G})}{\text{std}(r_{1:G})}$$

where $r_i \in \{0, 1\}$ is the binary correctness reward. The GRPO objective is

$$\mathcal{J}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(\hat{\rho}_{i,t} \hat{A}_i, \text{clip}(\hat{\rho}_{i,t}, 1-\epsilon, 1+\epsilon) \hat{A}_i \right) \right]$$

where $\hat{\rho}_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t}) / \pi_{\text{ref}}(o_{i,t} \mid q, o_{i,<t})$. JustGRPO differs only in that rollouts o_i are generated left-to-right.

Coverage collapse is measured by the expected entropy of the rollout prefix distribution at each position k :

$$\mathcal{H}_k = H(\{o_{i,1:k}\}_{i=1}^G).$$

Under arbitrary-order dLLM rollouts, $\mathcal{H}_k$ drops sharply by position 3; under JustGRPO it remains high throughout.

Learning or Inference Procedure

1. Sample a math/code problem q from the training set.
2. Generate $G = 8$ rollouts using **left-to-right decoding** with temperature 1.0.
3. Score each rollout for correctness: $r_i \in \{0, 1\}$.
4. Compute GRPO advantages $\hat{A}_i$ from the group reward distribution.
5. Update θ via the clipped surrogate loss with KL penalty β .
6. At **inference**, use native parallel decoding (arbitrary order).

What the Guarantee Says

The paper does not provide formal convergence guarantees, but proves empirically that coverage collapse is absent under left-to-right constraint and universal under arbitrary-order generation across all tested model families. The key invariant is: $\mathcal{H}_k^{\text{JustGRPO}} \geq c \cdot k$ for a positive constant c throughout training, while $\mathcal{H}_k^{\text{arbitrary}} \rightarrow 0$ within a few gradient steps.

Experimental Findings

Method	GSM8K	MATH-500
ESPO	82.3%	39.0%
GDPO	82.8%	39.6%
JustGRPO	89.1%	45.1%

On AIME 2024 (competition-level math), JustGRPO reaches 23.3% vs. GDPO’s 17.8%. Closed-source dLLM baselines without RL lag further behind. JustGRPO uses the same base model (MDLM) and identical training compute as the comparison methods.

Ablations and Interpretation

Order constraint at inference: Forcing left-to-right at inference *reduces* accuracy by 1.8 points, confirming the constraint should apply to training only.

Group size G : Performance is robust to $G \in \{4, 8, 16\}$; the coverage collapse under arbitrary order worsens with larger G because degenerate prefixes dominate faster.

Base model: Results replicate on MDLM, PLAID, and a dLLM based on LLaMA-3-8B. The effect is architectural, not model-specific.

Task type: Creative writing shows minimal gap (0.3 points), confirming the effect is specific to tasks with verifiable, structured correct answers where high-uncertainty tokens are decision-critical.

Reference

Zanlin Ni, Shenzhi Wang, Yang Yue, Tianyu Yu, Weilin Zhao, Yeguo Hua, Tianyi Chen, Jun Song, Cheng Yu, Bo Zheng, Gao Huang. **The Flexibility Trap: Rethinking the Value of Arbitrary Order in Diffusion Language Models.** arXiv:2601.15165, January 2026. <https://arxiv.org/abs/2601.15165>